

Project Design Phase-II

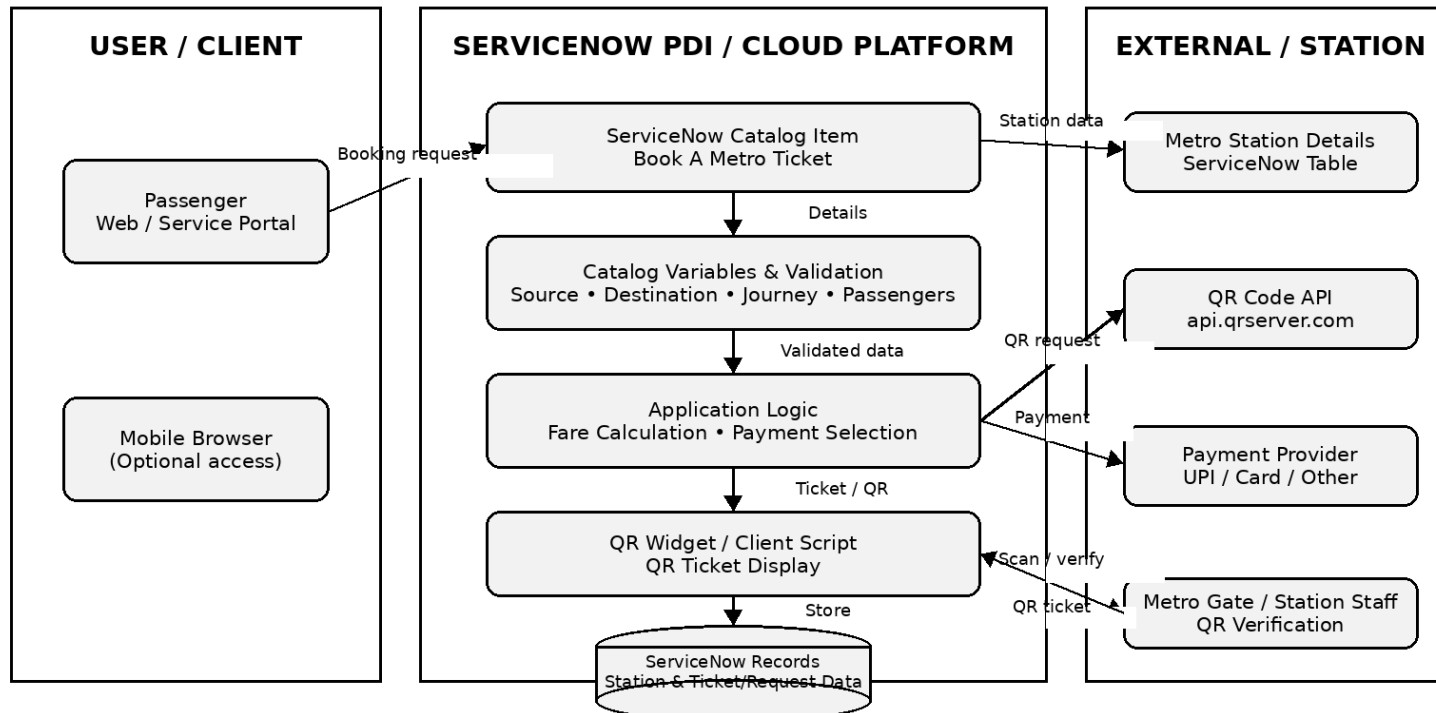
Technology Stack (Architecture & Stack)

Date	5 September 2026
Team ID	SWUID2026191970
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

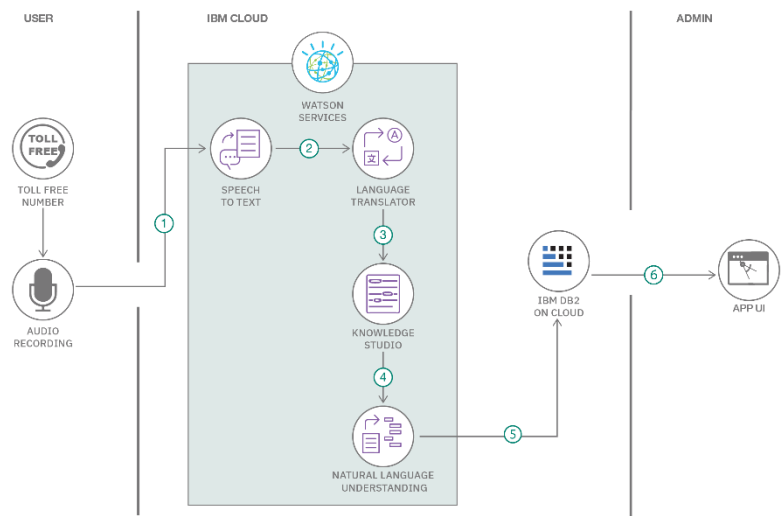
Metro Ticket Generating System - Technical Architecture



Infrastructure demarcation: user/client → ServiceNow cloud/PDI → external QR/payment and station-gate interactions.

Example: Metro Ticket Booking and QR Ticket Generation

Project-specific architecture prepared for the Metro Ticket Generating System.



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface used by passengers to book metro tickets and view the QR ticket.	ServiceNow Service Portal / Catalog UI, HTML, CSS, JavaScript
2	Application Logic-1	Ticket booking form processing and field validation.	ServiceNow Catalog Item & Catalog Client Script
3	Application Logic-2	Automatic fare calculation based on journey and passenger details.	ServiceNow Client Script / JavaScript
4	Application Logic-3	QR ticket display and ticket interaction.	ServiceNow Portal Widget, AngularJS, JavaScript
5	Database	Stores metro station details and ticket/request information.	ServiceNow platform tables / database
6	Cloud Database	Cloud-hosted data storage provided by the ServiceNow platform.	ServiceNow Cloud Database

7	File Storage	No separate file-storage service is required for the core solution.	ServiceNow platform storage / configuration records
8	External API-1	Generates the QR image used for the digital metro ticket.	QR Server API (api.qrserver.com)
9	External API-2	Payment processing interface represented by the selected payment mode.	UPI / Card / Other payment interface
10	Machine Learning Model	No machine learning model is required for the current project.	Not Applicable
11	Infrastructure (Server / Cloud)	Hosts the catalog item, scripts, widget, tables and project configuration.	ServiceNow PDI / Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The solution uses standard web scripting and ServiceNow portal technologies; no separate open-source application framework is required.	HTML, CSS, JavaScript; ServiceNow Portal/Widget framework
2	Security	Access to project records is controlled using ServiceNow access-control mechanisms. The solution does not require storing payment credentials in the custom tables.	ServiceNow ACLs / roles / platform security
3	Scalable Architecture	The solution separates the user interface, catalog/application logic, QR integration and data storage components, allowing additional stations and ticket records to be supported.	ServiceNow platform architecture
4	Availability	The application is delivered through the ServiceNow platform and is available when the ServiceNow instance and required services are operational.	ServiceNow cloud/PDI platform
5	Performance	The design keeps booking logic lightweight, uses client-side validation/calculation where appropriate, and generates the QR image through an external API.	JavaScript, ServiceNow Catalog, QR API

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>